$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{0^+}{}^\dagger & -\frac{3 k^2 \overset{(4)}{\kappa}_1}{2} & 0 \\
\mathcal{K}^{\#1}_{0^-}{}^{\dagger\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{K}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{1^-}{}_{\alpha} & \mathcal{K}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{K}^{\#1}_{1^+}{}^{\dagger\alpha\beta} & -k^2 \overset{(4)}{\kappa}_3 & -\frac{k^2 \overset{(4)}{\kappa}_3}{\sqrt{2}} & 0 & 0 \\
\mathcal{K}^{\#2}_{1^+}{}^{\dagger\alpha\beta} & -\frac{k^2 \overset{(4)}{\kappa}_3}{\sqrt{2}} & -\frac{k^2 \overset{(4)}{\kappa}_3}{2} & 0 & 0 \\
\mathcal{K}^{\#1}_{1^-}{}^{\dagger\alpha} & 0 & 0 & 0 & 0 \\
\mathcal{K}^{\#2}_{1^-}{}^{\dagger\alpha} & 0 & 0 & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{2^+}{}^{\dagger\alpha\beta} & 0 & 0 \\
\mathcal{K}^{\#1}_{2^-}{}^{\dagger\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{0^+}{}^\dagger & -\frac{2}{3 k^2 \overset{(4)}{\kappa}_1} & 0 \\
\mathcal{J}^{\#1}_{0^-}{}^{\dagger\alpha\beta\chi} & 0 & 0
\end{array}
$$

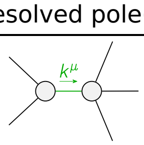
$$
\begin{array}{c|cccc}
 & \mathcal{J}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{1^-}{}_{\alpha} & \mathcal{J}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{J}^{\#1}_{1^+}{}^{\dagger\alpha\beta} & -\frac{4}{9 k^2 \overset{(4)}{\kappa}_3} & -\frac{2\sqrt{2}}{9 k^2 \overset{(4)}{\kappa}_3} & 0 & 0 \\
\mathcal{J}^{\#2}_{1^+}{}^{\dagger\alpha\beta} & -\frac{2\sqrt{2}}{9 k^2 \overset{(4)}{\kappa}_3} & -\frac{2}{9 k^2 \overset{(4)}{\kappa}_3} & 0 & 0 \\
\mathcal{J}^{\#1}_{1^-}{}^{\dagger\alpha} & 0 & 0 & 0 & 0 \\
\mathcal{J}^{\#2}_{1^-}{}^{\dagger\alpha} & 0 & 0 & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{2^+}{}^{\dagger\alpha\beta} & 0 & 0 \\
\mathcal{J}^{\#1}_{2^-}{}^{\dagger\alpha\beta\chi} & 0 & 0
\end{array}
$$

Lagrangian

$$\overset{(4)}{\kappa}_1 \partial_\beta \mathcal{K}^{\delta}{}_{\chi\delta} \partial^\chi \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^{\delta} - 2\, \overset{(4)}{\kappa}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^{\delta} - \overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^{\delta} - \frac{1}{2} \overset{(4)}{\kappa}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}{}_{\beta\chi}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\ \mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}^{\#1\alpha\beta\chi}_{0^-} == 0$	1	$\partial_\delta\partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta\partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta\partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta\partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta\partial^\delta \mathcal{J}^{\beta\alpha\chi} ==$ $\partial_\delta\partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta\partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta\partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta\partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta \mathcal{J}^{\chi\alpha\beta}$
$\mathcal{J}^{\#2\alpha}_{1^-} == 0$	3	$\partial_\chi\partial_\beta \mathcal{J}^{\beta\alpha\chi} == 0$
$\mathcal{J}^{\#1\alpha}_{1^-} == 0$	3	$\partial_\chi\partial^\alpha \mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi \mathcal{J}^{\beta\alpha}{}_{\beta} == \partial_\chi\partial_\beta \mathcal{J}^{\beta\alpha\chi}$
$\mathcal{J}^{\#1\alpha\beta}_{1^+} == \mathcal{J}^{\#2\alpha\beta}_{1^+}$	3	$3\,\partial_\delta\partial_\chi\partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\delta\partial_\chi \mathcal{J}^{\beta\alpha\chi} + 2\,\partial_\delta\partial^\delta\partial_\chi \mathcal{J}^{\chi\alpha\beta} == 3\,\partial_\delta\partial_\chi\partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\delta\partial_\chi \mathcal{J}^{\alpha\beta\chi}$
$\mathcal{J}^{\#1\alpha\beta\chi}_{2^-} == 0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha \mathcal{J}^{\delta\beta}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta \mathcal{J}^{\alpha\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi \mathcal{J}^{\beta\alpha\delta} +$ $4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3\ \eta^{\beta\chi}\ \partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha \mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\ \eta^{\alpha\chi}\ \partial_\phi\partial^\phi\partial_\epsilon\partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3\ \eta^{\beta\chi}\ \partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon \mathcal{J}^{\delta\alpha}{}_{\delta} ==$ $3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta \mathcal{J}^{\delta\alpha}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi \mathcal{J}^{\alpha\beta\delta} +$ $2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3\ \eta^{\alpha\chi}\ \partial_\phi\partial^\phi\partial_\epsilon\partial^\beta \mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\ \eta^{\beta\chi}\ \partial_\phi\partial^\phi\partial_\epsilon\partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3\ \eta^{\alpha\chi}\ \partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon \mathcal{J}^{\delta\beta}{}_{\delta}$
$\mathcal{J}^{\#1\alpha\beta}_{2^+} == 0$	5	$3\,\partial_\delta\partial_\chi\partial^\alpha \mathcal{J}^{\chi\beta\delta} + 3\,\partial_\delta\partial_\chi\partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2\ \eta^{\alpha\beta}\ \partial_\epsilon\partial^\epsilon\partial_\delta \mathcal{J}^{\chi}{}_{\chi}{}^{\delta} == 2\,\partial_\delta\partial^\beta\partial^\alpha \mathcal{J}^{\chi}{}_{\chi}{}^{\delta} + 3\,(\partial_\delta\partial^\delta\partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\partial_\chi \mathcal{J}^{\beta\alpha\chi})$
Total # constraint(s):	20	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$-\frac{1}{\overset{(4)}{\kappa}_3}$
Resolved unitarity condition(s):	$\overset{(4)}{\kappa}_3 < 0$		